



Energetic Variational Neural Networks for Multiscale Problems

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ABSTRACT

Multiscale problems featuring sharp interfaces arise broadly in science and engineering, including phase-field models, pattern formation, and liquid crystal dynamics. Accurate numerical simulation of these problems using neural networks is challenging due to three fundamental difficulties: (1) *sharp spatial structures* that trigger spectral bias in neural networks, (2) *ill-conditioning* of the optimization landscape at small parameters, and (3) the need to guarantee energy dissipation throughout time evolution.

In this work, we develop a comprehensive neural network approach combining three complementary methodological innovations. We employ **Physics-Informed Neural Networks (PINN)** and **Energetic Variational Neural Networks (EVNN)**, which discretizes gradient-flow dynamics variationally to guarantee energy dissipation at every time step. We implement **two-scale feature embedding**, augmenting network inputs with rescaled coordinates to reduce spectral bias and resolve sharp $O(\varepsilon)$ features. We employ **ε -continuation training**, progressively decreasing the small parameter to avoid ill-conditioning.

We validate the combined approach on one- and two-dimensional Allen–Cahn equations, benchmarking PINN and EVNN with both Fourier feature and two-scale embeddings. Results confirm that two-scale features successfully resolve sharp interfaces, that ε -continuation enables convergence at small target ε values, and that EVNN preserves energy dissipation by construction. Application to three-dimensional liquid crystal shells demonstrates that the proposed methodology qualitatively reproduces topological defect structures with sharp localization and physically consistent director-field organization.

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1 Introduction

Phase-field models, pattern formation, and liquid crystal dynamics all give rise to partial differential equations (PDEs) whose solutions develop sharp interfaces—thin transition layers of width $O(\varepsilon)$ across which the solution jumps between distinct states. As $\varepsilon \rightarrow 0$, resolving these layers becomes increasingly expensive for standard numerical methods. A canonical example is the Allen–Cahn equation, a phase-field model in which the solution transitions sharply between two stable states across an interfacial layer of width $O(\varepsilon)$, where $\varepsilon \ll 1$ is a small parameter. As $\varepsilon \rightarrow 0$, the interface becomes increasingly sharp, and standard numerical methods must resolve fine-scale structure at significant computational cost. Similar multiscale phenomena appear in liquid crystal systems, where topological defect cores exhibit sharp transitions in the order parameter across length scales of $O(\varepsilon)$.

Neural network-based methods for solving PDEs have attracted significant interest due to their mesh-free nature and flexibility. However, applying these methods to multiscale problems with sharp interfaces introduces three fundamental challenges.

Spectral bias. Standard feedforward neural networks with coordinate inputs exhibit spectral bias: they preferentially learn low-frequency features and struggle to represent high-frequency or sharply varying solution components [Zhuang et al., 2025]. For problems with sharp interfaces, this leads to blurred approximations that fail to resolve the true solution structure. **Ill-conditioning.** The presence of multiple length scales—macroscopic geometry at $O(1)$ and interfacial width at $O(\varepsilon)$ —creates severe ill-conditioning in the optimization landscape. Training directly at small ε leads to slow convergence and optimizer stagnation. **Energy dissipation.** Many physical systems satisfy a fundamental energy-dissipation law: the free energy must decrease monotonically with time. Standard residual-based neural network training does not explicitly encode this thermodynamic structure, potentially leading to unphysical solutions.

To address these challenges, we develop and validate a combination of three complementary innovations: two-scale feature embedding to resolve sharp interfaces, ε -continuation training to handle ill-conditioning, and Energetic Variational Neural Networks (EVNN) to guarantee energy dissipation by construction. We apply these methods to the Allen–Cahn equation as a model problem and to liquid crystal shells as a physically rich application.

2 Methodology

2.1 Physics-Informed Neural Networks

Physics-informed neural networks (PINN) are a class of neural-network-based methods for solving PDEs by embedding physical laws directly into the training objective. Instead of learning solely from labeled data, PINN approximate the solution of a PDE using a neural network and enforce the governing equations, as well as initial and boundary conditions, through the loss function.

Consider a general time-dependent PDE of the form

$$\mathcal{N}[u(x, t)] = 0, \quad (x, t) \in \Omega \times (0, T], \quad (1)$$

where $u(x, t)$ denotes the unknown solution and $\mathcal{N}$ is a (possibly nonlinear) differential operator. In the PINN framework, the solution is approximated by a neural network $u(x, t) \approx u_{\text{NN}}(x, t; \Theta)$ with trainable parameters Θ . Using automatic differentiation, the physics-informed residual is

$$R(x, t; \Theta) = \mathcal{N}[u_{\text{NN}}(x, t; \Theta)], \quad (2)$$

and the network is trained by minimizing the composite loss

$$\mathcal{L}_{\text{PINN}} = \frac{1}{N_f} \sum_{i=1}^{N_f} |R(x_i, t_i; \Theta)|^2 + \frac{\lambda_{\text{IC}}}{N_0} \sum_{i=1}^{N_0} |u_{\text{NN}}(x_i, 0; \Theta) - u_0(x_i)|^2 + \frac{\lambda_{\text{BC}}}{N_b} \sum_{i=1}^{N_b} |u_{\text{NN}}(x_i, t_i; \Theta) - g(x_i, t_i)|^2. \quad (3)$$

PINN provide a mesh-free and flexible framework, but since training is based on residual minimization, they do not explicitly encode structural properties such as energy dissipation. In addition, training is often hampered by spectral bias: PINN tend to perform well for smooth, low-frequency solutions but struggle to capture high-frequency or sharply varying components.

2.2 Energetic Variational Neural Networks

Energetic Variational Neural Networks (EVNN) are a structure-preserving approach to discretizing gradient flows [Hu et al., 2024]. Rather than minimizing PDE residuals, EVNN directly discretizes the underlying energy-dissipation structure. For

an L^2 -gradient flow with free energy $\mathcal{F}[\varphi]$ and mobility $\eta(\varphi)$, the energy-dissipation law takes the form

$$\frac{d}{dt}\mathcal{F}(\varphi) = - \int_{\Omega} \eta(\varphi) |\varphi_t|^2 dx. \quad (4)$$

Given the solution $\varphi_{\text{NN}}(\mathbf{x}; \Theta^n)$ at time t^n , the network parameters Θ^{n+1} at the next time step $t^{n+1} = t^n + \tau$ are obtained by solving

$$\Theta^{n+1} = \arg \min_{\Theta} \left(\frac{1}{2\tau} \int_{\Omega} \eta^n |\varphi_{\text{NN}}(\mathbf{x}; \Theta) - \varphi_{\text{NN}}(\mathbf{x}; \Theta^n)|^2 d\mathbf{x} + \mathcal{F}[\varphi_{\text{NN}}(\mathbf{x}; \Theta)] \right). \quad (5)$$

The variational structure guarantees monotonic energy decay: $\mathcal{F}[\varphi_{\text{NN}}(\mathbf{x}; \Theta^{n+1})] \leq \mathcal{F}[\varphi_{\text{NN}}(\mathbf{x}; \Theta^n)]$ by construction, independent of optimization details.

2.3 Fourier Feature Embedding

As a baseline enhancement for handling periodic spatial structure, we consider a Fourier feature embedding for the spatial variable. For a domain with period L and prescribed number of modes K , the Fourier feature map is

$$\phi_{\text{FF}}(x) = \left[\cos\left(\frac{2\pi kx}{L}\right), \sin\left(\frac{2\pi kx}{L}\right) \right]_{k=1}^K \in \mathbb{R}^{2K}. \quad (6)$$

The features are fixed (non-trainable) and encode the periodic structure of the domain exactly. While the Fourier embedding effectively resolves periodic spatial structure, it does not encode information about the interfacial length scale $O(\varepsilon)$ explicitly. As discussed in Section 4.6, this limits its effectiveness for problems with very small ε .

2.4 Two-Scale Feature Embedding

To resolve sharp interfaces at scale $O(\varepsilon)$, we introduce an ε -dependent two-scale feature embedding [Zhuang et al., 2025]:

$$\phi_{2s}(x; \varepsilon) = \begin{bmatrix} x \\ x/\sqrt{\varepsilon} \\ 1/\sqrt{\varepsilon} \end{bmatrix}. \quad (7)$$

The first component retains the macroscopic spatial variable. The second component $x/\sqrt{\varepsilon}$ rescales space to resolve interfacial variation at scale $O(\varepsilon)$. The constant $1/\sqrt{\varepsilon}$ provides scale-aware bias. This method requires only a simple modification to the network input and no changes to the architecture or training procedure beyond the ε -continuation strategy. Unlike the Fourier embedding, the two-scale map explicitly encodes the value of ε , making it directly adapted to the sharp interfacial structure of the problem.

2.5 ε -Continuation Training

Even with two-scale feature embedding, direct training at small target ε leads to an ill-conditioned optimization problem. We employ a progressive warm-starting approach [Zhuang et al., 2025], starting with a relatively large ε and progressively decreasing:

$$\varepsilon_{\text{large}} \longrightarrow \cdots \longrightarrow \varepsilon_{\text{target}} \quad (8)$$

At each stage, the network is trained to convergence and its parameters serve as warm-start initialization for the next stage. At large ε , the problem is well-conditioned and standard optimizers converge easily. As ε decreases, the network gradually learns to resolve finer scales, never facing a completely ill-conditioned problem from scratch.

3 Neural Network Approximation of Sharp Transitions

Approximating functions with sharp transitions is a fundamental challenge for neural networks. As illustrated in Figure 1, many PDE solutions exhibit sharp transitions across an interfacial layer of width $O(\varepsilon)$, where the solution transitions rapidly between two distinct states. As $\varepsilon \rightarrow 0$, the gradient within this layer grows as $O(1/\varepsilon)$, making it increasingly difficult for a standard network with raw coordinate inputs to represent.

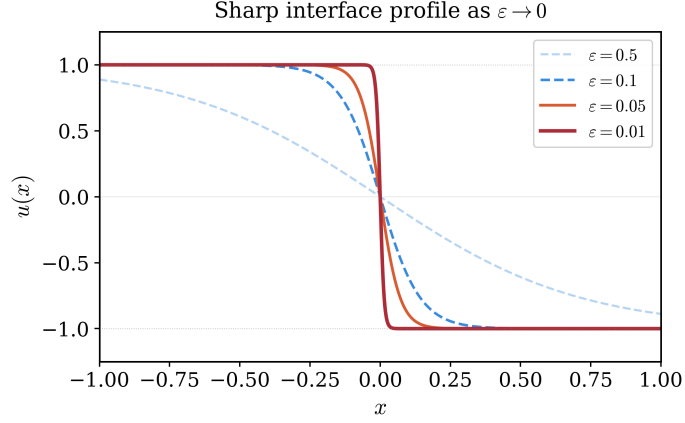


Figure 1: Sharp interface profiles $u(x)$ for decreasing values of ε . As $\varepsilon \rightarrow 0$, the transition between the two bulk states $u = \pm 1$ becomes increasingly localized, posing fundamental challenges for neural network approximation.

This difficulty is a consequence of spectral bias: multilayer perceptrons trained with gradient-based methods tend to learn low-frequency components of the solution first and struggle to represent high-frequency or sharply varying features. For problems with small ε , a network that has not been given explicit information about the interfacial scale will produce smoothed approximations that fail to capture the sharp transition.

The two-scale feature embedding (7) directly addresses this by providing the network with a rescaled coordinate $x/\sqrt{\varepsilon}$ that varies on the scale of the interface. Combined with the ε -continuation strategy, which conditions the optimizer by starting from a well-posed large- ε problem, the network can successfully resolve sharp interfaces even at $\varepsilon = 10^{-4}$.

4 1D Allen–Cahn

4.1 Problem Setup

We consider the one-dimensional Allen–Cahn equation on $[-1, 1]$ with periodic boundary conditions [Wight and Zhao, 2020]:

$$\begin{cases} u_t - \varepsilon^2 u_{xx} + 5(u^3 - u) = 0, & x \in [-1, 1], \quad t \in [0, 1], \\ u(0, x) = x^2 \cos(\pi x), & x \in [-1, 1], \\ u(t, -1) = u(t, 1), \quad u_x(t, -1) = u_x(t, 1), & t \in [0, 1]. \end{cases} \quad (9)$$

4.2 PINN: Fourier Feature Method

Within the PINN framework, the solution is approximated by $U_\theta(t, x)$ with the hard-constraint ansatz

$$U_\theta(t, x) = e^{-\varepsilon^2 t} u_0(x) + t N_\theta(t, x), \quad (10)$$

ensuring $U_\theta(0, x) = u_0(x)$ exactly. The network input uses the Fourier feature map (6) with $K = 20$. A fully connected network with three hidden layers of width 128 and tanh activations is trained using Adam at learning rate 10^{-3} without any continuation strategy.

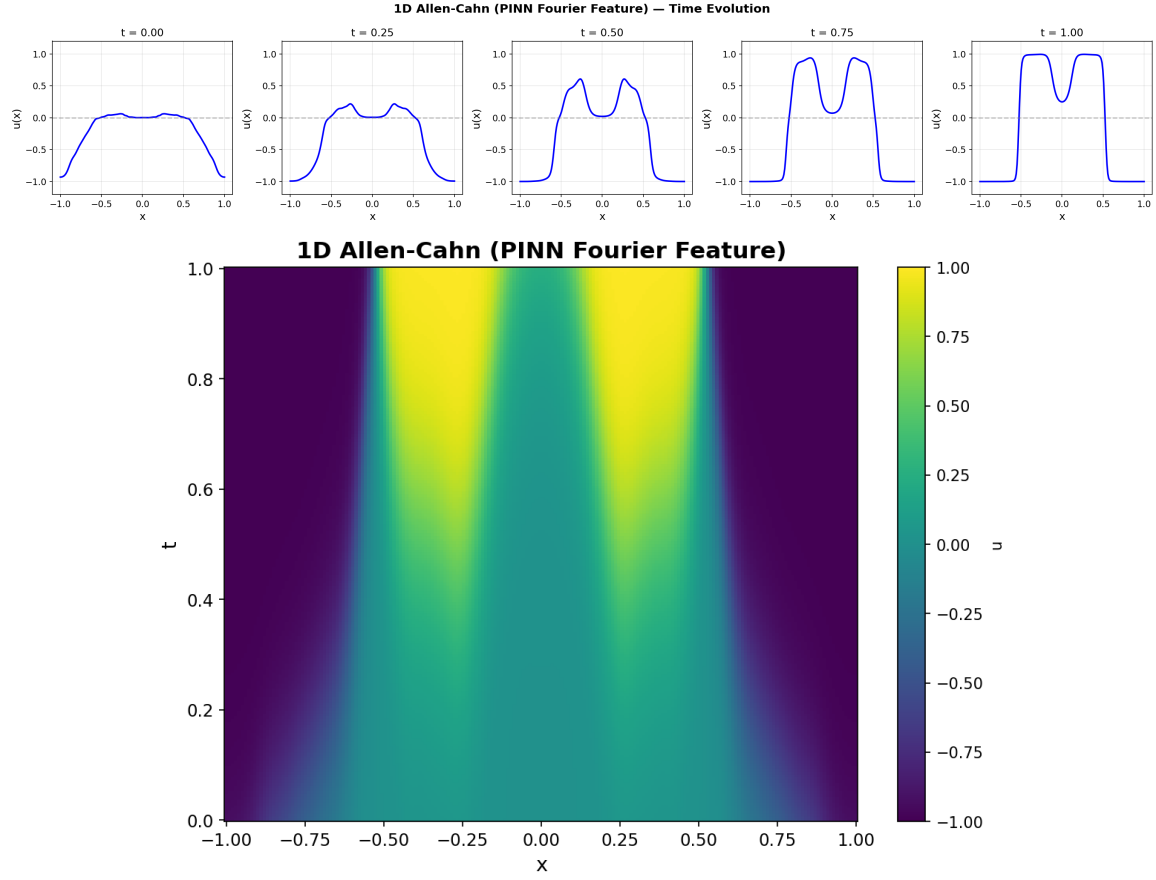


Figure 2: PINN Fourier feature method for the 1D Allen-Cahn equation with $\varepsilon^2 = 10^{-4}$. *Top:* solution profiles at $t = 0, 0.25, 0.5, 0.75, 1.0$. *Bottom:* space-time heatmap.

4.3 PINN: Two-Scale Method

We replace the Fourier feature map with the two-scale embedding (7), apply the ε -continuation strategy, and train with AdamW using weight decay 10^{-4} .

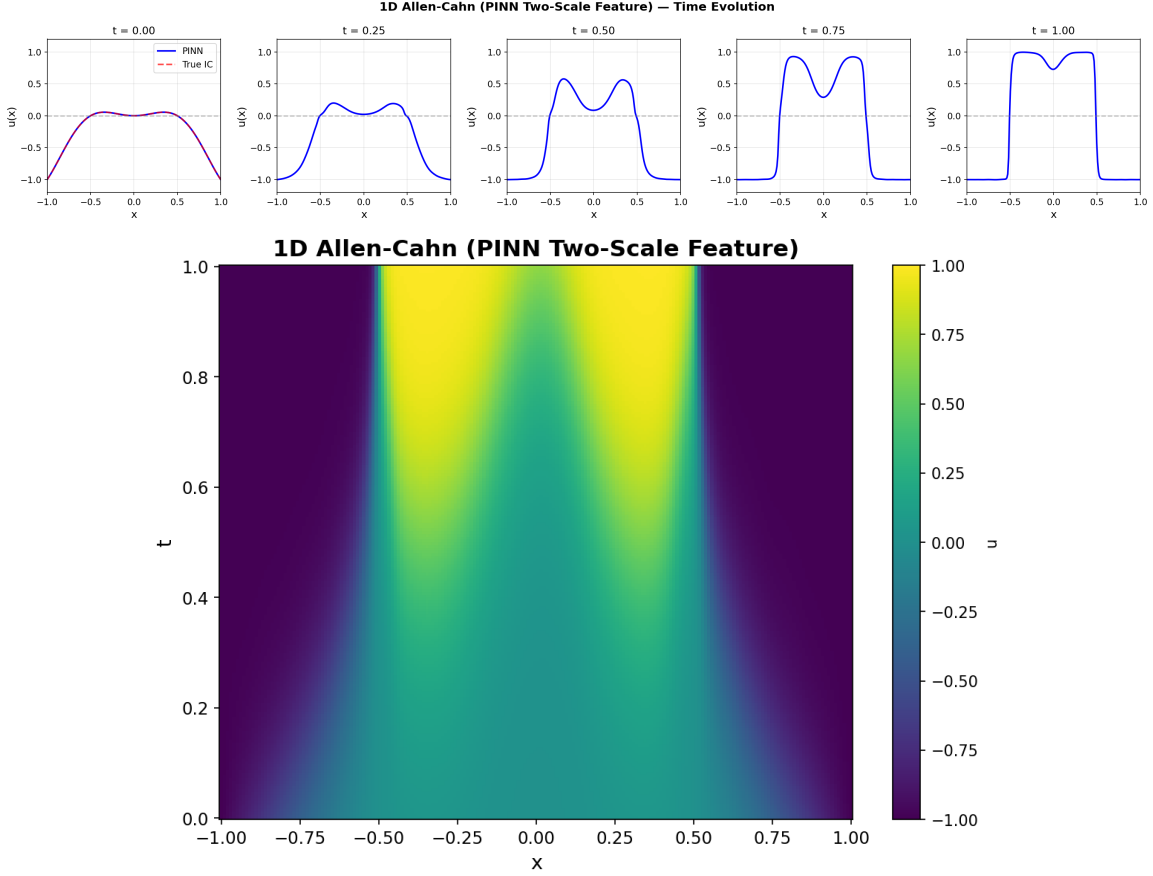


Figure 3: PINN two-scale method for the 1D Allen–Cahn equation with $\varepsilon^2 = 10^{-4}$. *Top:* solution profiles at $t = 0, 0.25, 0.5, 0.75, 1.0$. *Bottom:* space–time heatmap. The sharp interfacial layers are well resolved at all times.

4.4 EVNN: Fourier Feature Method

Within the EVNN framework, time is handled explicitly through the variational time-stepping scheme (5); the network represents the spatial solution $u(x; \Theta^n)$ at each discrete time level. The Fourier feature map (6) with $K = 20$ is used as input. The spatial domain is discretized with $N_x = 256$ uniform points; spatial derivatives are approximated by second-order central finite differences with periodic wrap-around. The free energy functional is

$$\mathcal{F}[u] = \int_{-1}^1 \left(\frac{\varepsilon^2}{2} |u_x|^2 + \frac{5}{4} (u^2 - 1)^2 \right) dx, \quad (11)$$

with time step $\tau = 0.002$ and 120 Adam iterations per step at learning rate 3×10^{-4} .

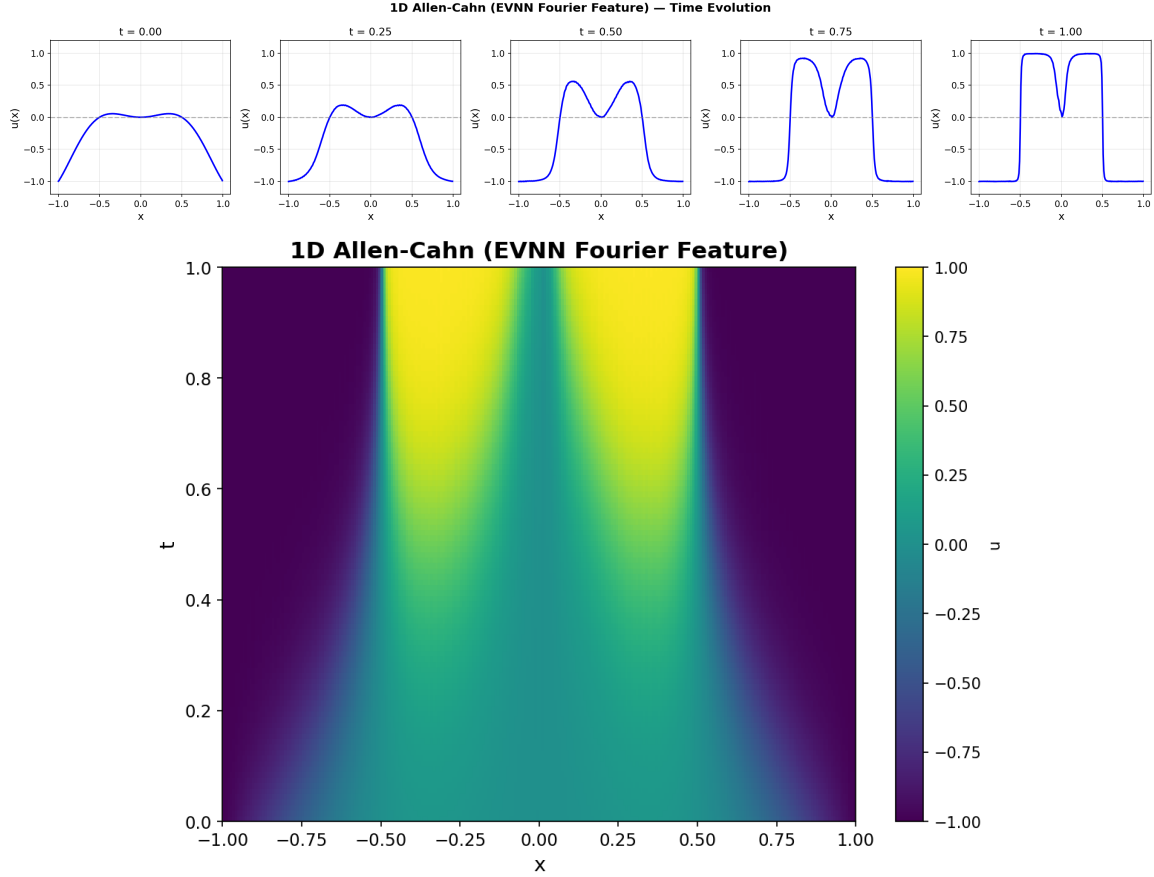


Figure 4: EVNN Fourier feature method for the 1D Allen-Cahn equation with $\varepsilon^2 = 10^{-4}$. *Top:* solution profiles at $t = 0, 0.25, 0.5, 0.75, 1.0$. *Bottom:* space-time heatmap.

4.5 EVNN: Two-Scale Method

The two-scale feature map (7) replaces the Fourier embedding, with all other aspects of the EVNN formulation identical to Section 4.4. This encodes both the macroscopic coordinate and the rescaled coordinate $x/\sqrt{\varepsilon}$, directly adapted to the interfacial width $O(\varepsilon)$.

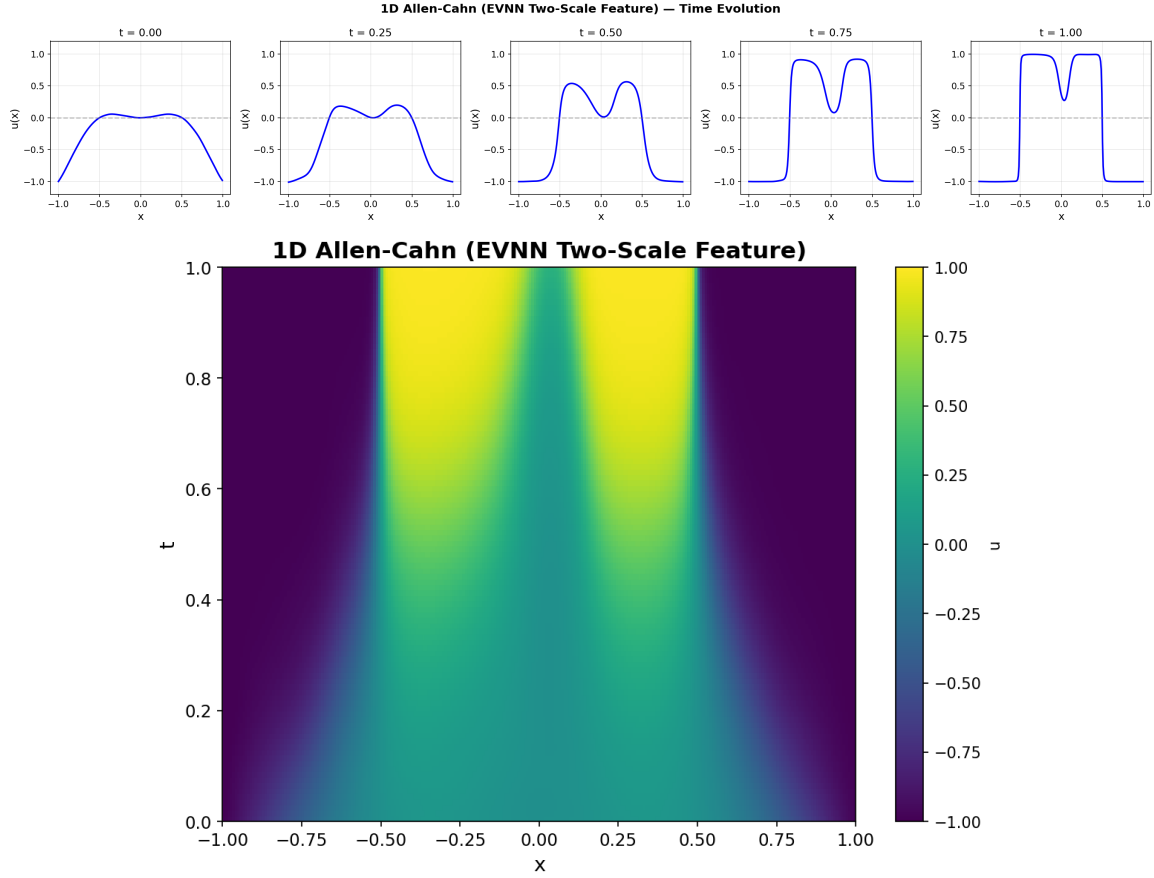


Figure 5: EVNN two-scale method for the 1D Allen–Cahn equation with $\varepsilon^2 = 10^{-4}$. *Top:* solution profiles at $t = 0, 0.25, 0.5, 0.75, 1.0$. *Bottom:* space–time heatmap.

4.6 Discussion: Why Two-Scale for Downstream Problems

Both Fourier and two-scale embeddings successfully resolve the sharp interface in the 1D Allen–Cahn problem at $\varepsilon^2 = 10^{-4}$. At this scale, the periodic structure of the domain is the dominant feature, and the fixed Fourier basis is sufficient to represent the solution accurately. The 2D Allen–Cahn benchmark in Section 5.4 reinforces this picture: the Fourier-feature EVNN attains lower pointwise errors against the FEM reference than the two-scale variant (Table 1). On these two model problems alone, the Fourier embedding would be the natural default choice.

The choice changes for the liquid crystal shell problem in Section 6. There $\varepsilon^2 \sim 10^{-4}$ on a curved, non-periodic domain, and the defect cores have $O(1/\varepsilon)$ -sized gradients localized on length scales much smaller than the macroscopic geometry. In

our experiments, the Fourier-feature method failed to produce a usable solution in this regime: the fixed periodic basis does not encode the interfacial scale, and training did not converge to a sharply resolved director field. The two-scale embedding, by contrast, exposes the rescaled coordinate $x/\sqrt{\varepsilon}$ to the network and—combined with ε -continuation to avoid ill-conditioning at the target ε —does converge.

We therefore use the two-scale method with ε -continuation for the liquid crystal problem out of practical necessity rather than uniform superiority. The takeaway is that the two embeddings are complementary: Fourier is preferable when the domain is periodic and ε is moderate, while two-scale is necessary when ε is small and the geometry does not admit a natural periodic basis.

5 2D Allen–Cahn

5.1 Problem Setup

We consider the two-dimensional Allen–Cahn equation on $\Omega = (-1, 1)^2$ with a Dirichlet boundary condition and volume-preserving penalty [Hu et al., 2024]:

$$\begin{cases} \frac{\partial \varphi}{\partial t}(\mathbf{x}, t) = -\left(\frac{F'(\varphi)}{\varepsilon^2} - \Delta \varphi(\mathbf{x}, t) + 2W\left(\int_{\Omega} \varphi d\mathbf{x} - A\right)\right), & \mathbf{x} \in \Omega, \quad t > 0, \\ \varphi(\mathbf{x}, t) = -1, & \mathbf{x} \in \partial\Omega, \quad t > 0, \\ \varphi(\mathbf{x}, 0) = -\tanh\left(10\left(\sqrt{4x_1^2 + x_2^2} - 0.5\right)\right), & \mathbf{x} \in \Omega, \end{cases} \quad (12)$$

where $F(\varphi) = \frac{1}{4}(\varphi^2 - 1)^2$ is the double-well potential (distinct from the free energy functional $\mathcal{F}$ defined below), $\varepsilon^2 = 10^{-2}$, volume target $A = -(4 - \pi r^2) + \pi r^2$ with $r = 0.5$, and penalty weight $W = 1000$. The system is an L^2 -gradient flow of the energy

$$\mathcal{F}[\varphi] = \int_{\Omega} \frac{1}{2} |\nabla \varphi|^2 + \frac{1}{4\varepsilon^2} (\varphi^2 - 1)^2 d\mathbf{x} + W \left(\int_{\Omega} \varphi d\mathbf{x} - A \right)^2. \quad (13)$$

Note that we do not apply PINN to the 2D Allen–Cahn problem. The volume-preserving penalty term with weight $W = 1000$ introduces a global integral constraint into the loss function, which significantly complicates residual-based training and makes it difficult for PINN to maintain the correct phase geometry throughout the evolution. Furthermore, this problem is specifically designed as a benchmark for the EVNN framework [Hu et al., 2024], and we follow the original setup accordingly.

5.2 EVNN: Fourier Feature Method

In the 2D setting, the Fourier feature embedding benefits from encoding the periodic spatial structure of the domain directly, without requiring knowledge of ε . This makes it well-suited for the Allen–Cahn problem where the domain periodicity is a dominant feature. By contrast, the two-scale method introduces additional approximation error during the continuation warm-up phase, particularly in fitting the initial condition at the target ε . Nevertheless, both methods successfully capture the qualitative interface dynamics and preserve monotonic energy dissipation throughout the simulation.

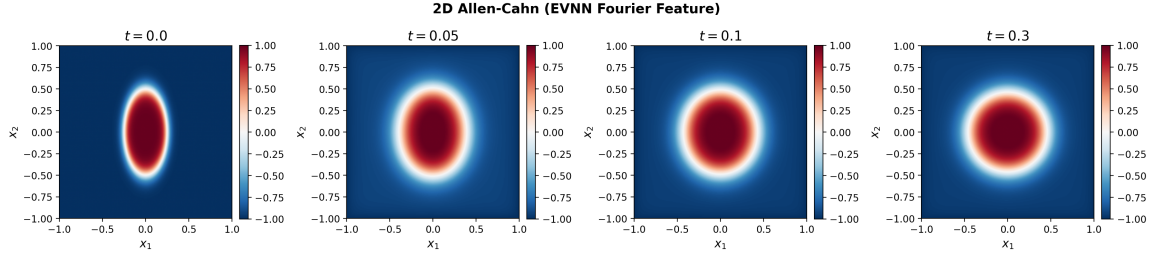


Figure 6: EVNN Fourier-feature method for the 2D Allen-Cahn equation. Solution snapshots $\varphi(\mathbf{x}, t^n)$ at $t = 0, 0.05, 0.1, 0.3$.

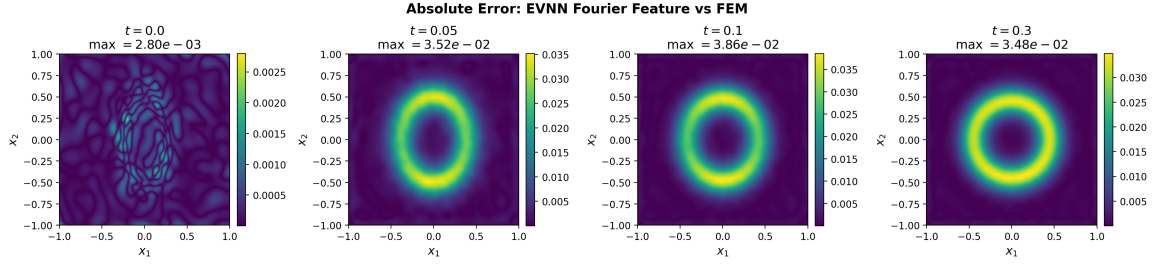


Figure 7: Pointwise absolute error between EVNN Fourier-feature solution and FEM reference.

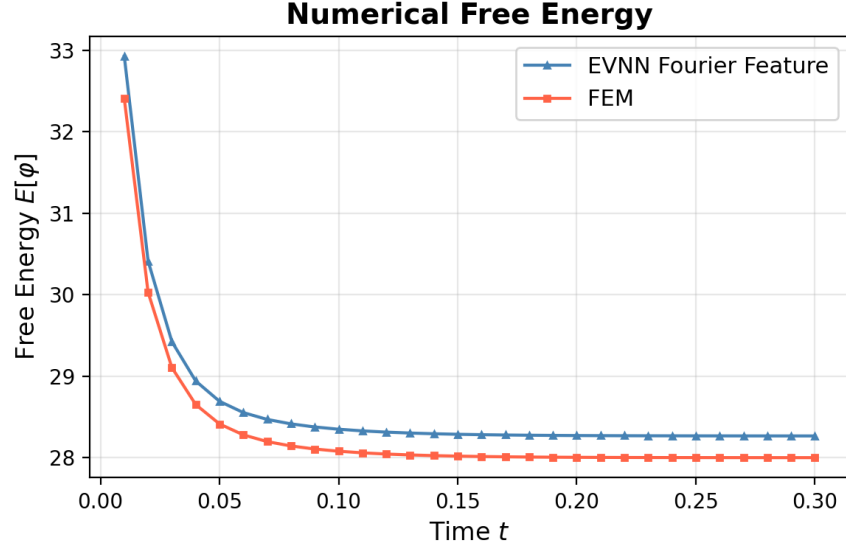


Figure 8: Discrete free energy $\mathcal{F}(t^n)$ for the EVNN Fourier-feature method compared with FEM reference. The EVNN energy decreases monotonically.

5.3 EVNN: Two-Scale Method with Continuation

The two-scale feature map applied to 2D is

$$\phi_{2s}(\mathbf{x}; \varepsilon) = \begin{bmatrix} x_1 \\ x_1/\sqrt{\varepsilon} \\ x_2 \\ x_2/\sqrt{\varepsilon} \\ 1/\sqrt{\varepsilon} \end{bmatrix}. \quad (14)$$

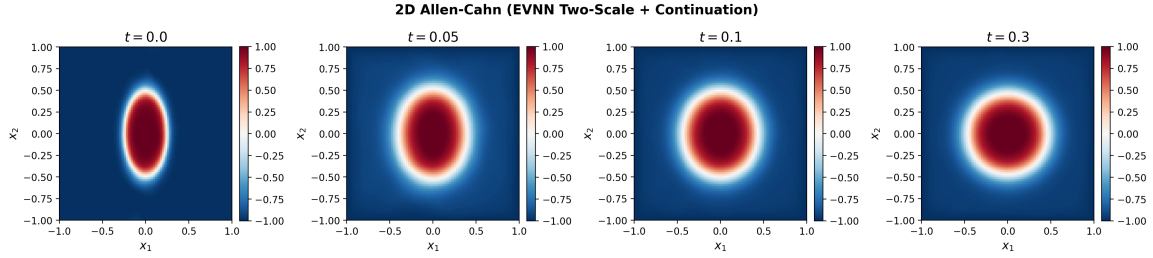


Figure 9: EVNN two-scale method with continuation for the 2D Allen–Cahn equation. Solution snapshots at $t = 0, 0.05, 0.1, 0.3$.

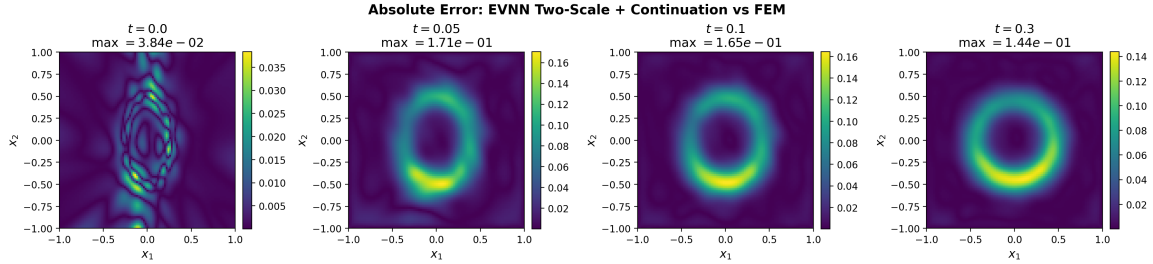


Figure 10: Pointwise absolute error between EVNN two-scale solution and FEM reference.

5.4 Comparison

Table 1 summarizes the maximum absolute errors and steady-state energies for both EVNN variants. The Fourier-feature method achieves lower errors in the 2D Allen–Cahn setting, while both methods preserve the monotonic energy-dissipation structure. On this benchmark alone, the Fourier embedding is the more accurate choice. However, this advantage does not transfer to the liquid crystal problem in

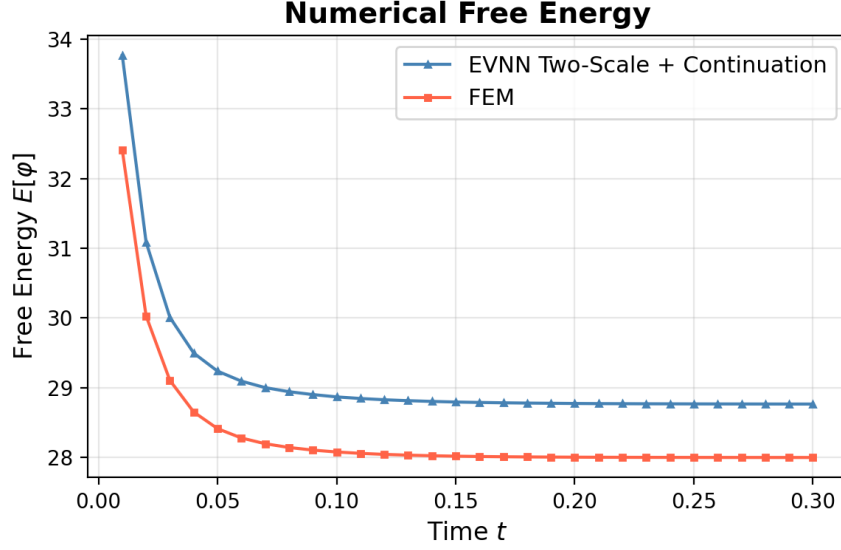


Figure 11: Discrete free energy $\mathcal{F}(t^n)$ for the EVNN two-scale method with continuation compared with FEM reference.

Section 6: as discussed in Section 4.6, the Fourier-feature EVNN failed to converge to a sharply resolved director field on the LC shell, where ε is much smaller and the domain is curved and non-periodic. The two-scale method with ε -continuation, while less accurate on this 2D benchmark, is the variant we are able to actually run on the LC problem, which is why it is used in Section 6.

Table 1: Comparison of EVNN Fourier-feature and two-scale methods for the 2D Allen–Cahn equation ($\varepsilon^2 = 10^{-2}$). The FEM solution is used as the reference, so its error entries are not applicable.

Method	Max absolute error vs. FEM				Steady-state energy $\mathcal{F}(T)$
	$t = 0$	$t = 0.05$	$t = 0.1$	$t = 0.3$	
FEM (reference)	—	—	—	—	≈ 28.0
EVNN Fourier	2.80×10^{-3}	3.52×10^{-2}	3.86×10^{-2}	3.48×10^{-2}	≈ 28.2
EVNN Two-scale	3.84×10^{-2}	1.71×10^{-1}	1.65×10^{-1}	1.44×10^{-1}	≈ 28.8

6 Application: Liquid Crystal Shells

6.1 Why Liquid Crystal Shells Matter

Liquid crystal shells are essential model systems for studying topological phenomena on curved surfaces. The Poincaré–Hopf theorem dictates that the total topological charge on a sphere must equal 2. When nematic liquid crystals are confined on a spherical shell with conflicting boundary conditions—homeotropic (normal) anchoring on the outer surface and degenerate planar (tangential) anchoring on the inner surface—these geometric constraints force the formation of topological defects. The structure and dynamics of these defects determine the overall optical and mechanical response of the LC shell, making accurate simulation critical for applications in photonics, soft robotics, and sensing [Noh et al., 2020].

6.2 Formation and Topology of Defects

In a spherical LC shell, the orientational order is described by a director field $\mathbf{n}(\mathbf{x})$, representing the local preferred orientation of liquid crystal molecules. Perfect alignment is geometrically impossible on a curved surface with competing boundary conditions. The total topological charge on each interface must sum to $+2$ by the Poincaré–Hopf theorem, forcing the formation of topological defects. Within these defects, the order parameter exhibits sharp transitions across length scales of $O(\varepsilon)$, where $\varepsilon^2 \sim 10^{-4}$, with defect cores characterized by $O(1/\varepsilon)$ -sized gradients. Capturing both the curved geometry and the small- ε defect structure requires a tensor-valued order parameter together with an energy functional that penalizes elastic distortion and enforces the boundary anchoring conditions. We introduce this formulation in the following section.

6.3 Mathematical Formulation

The state of a nematic liquid crystal is described by a symmetric, traceless 3×3 tensor $\mathbf{Q}(\mathbf{x})$, the Q-tensor order parameter. The Landau–de Gennes free energy is [Noh et al., 2020]:

$$\mathcal{F}[\mathbf{Q}] = \int_{\Omega} [f_{\text{bulk}}(\mathbf{Q}) + f_{\text{elastic}}(\mathbf{Q}, \nabla \mathbf{Q})] d\mathbf{x} + \int_{\partial\Omega} f_{\text{surface}}(\mathbf{Q}) dA, \quad (15)$$

where

$$f_{\text{bulk}}(\mathbf{Q}) = \frac{t}{2} \text{tr}(\mathbf{Q}^2) - \sqrt{6} \text{tr}(\mathbf{Q}^3) + \frac{1}{2} (\text{tr}(\mathbf{Q}^2))^2, \quad (16)$$

$$f_{\text{elastic}}(\mathbf{Q}, \nabla \mathbf{Q}) = \frac{\xi_R^2}{2} (\partial_{x_k} Q_{ij} \partial_{x_k} Q_{ij} + \eta \partial_{x_j} Q_{ij} \partial_{x_k} Q_{ik}), \quad (17)$$

with $\xi_R^2 = O(\varepsilon)$ the elastic correlation length, and surface energies enforcing boundary conditions at the inner and outer interfaces. Time evolution follows the gradient flow [Noh et al., 2020]:

$$\tilde{\mathbf{Q}}_t = -M(\mathbf{x}) \frac{\delta \mathcal{F}}{\delta \tilde{\mathbf{Q}}}, \quad \frac{d}{dt} \mathcal{F}[\tilde{\mathbf{Q}}] = - \int_{\Omega} M(\mathbf{x}) \tilde{\mathbf{Q}}_t \cdot \tilde{\mathbf{Q}}_t d\mathbf{x} \leq 0. \quad (18)$$

We consider shells confined in the annular domain $\Omega = B_1(\mathbf{0}) \setminus B_{\rho}(\mathbf{x}_c)$, with outer radius $R = 1$, inner radius $\rho = 0.7$, and offset $\mathbf{x}_c = (0, 0, 0.1)$

6.4 Results: Baseline vs. Proposed Method

The baseline approach uses EVNN without two-scale features or ε -continuation. As shown in Figure 12, this produces heavily blurred representations of the topological defect structures, failing to resolve the sharp director field organization at defect cores.

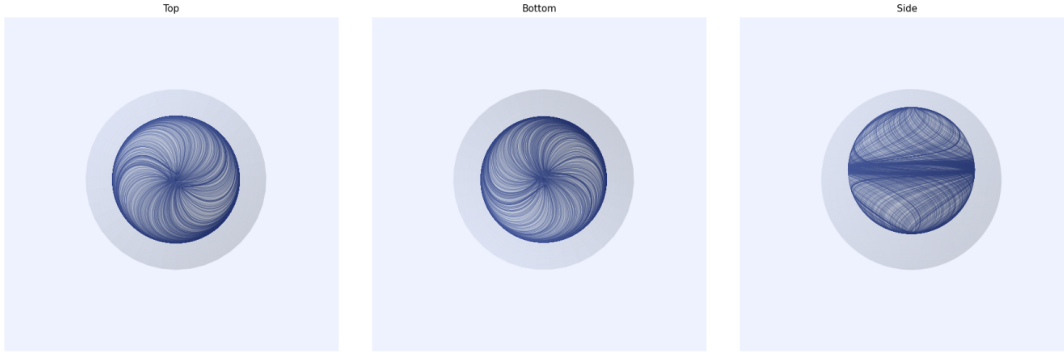


Figure 12: Baseline result using EVNN without two-scale features or ε -continuation. The defect structures are blurred and the director field streamlines are not clearly resolved.

Our proposed method—combining EVNN, two-scale features, and ε -continuation—produces results with dramatically improved clarity, as shown in Figure 13. The defect cores are resolved with sharp localization, and the director field streamlines are crisp and physically meaningful.

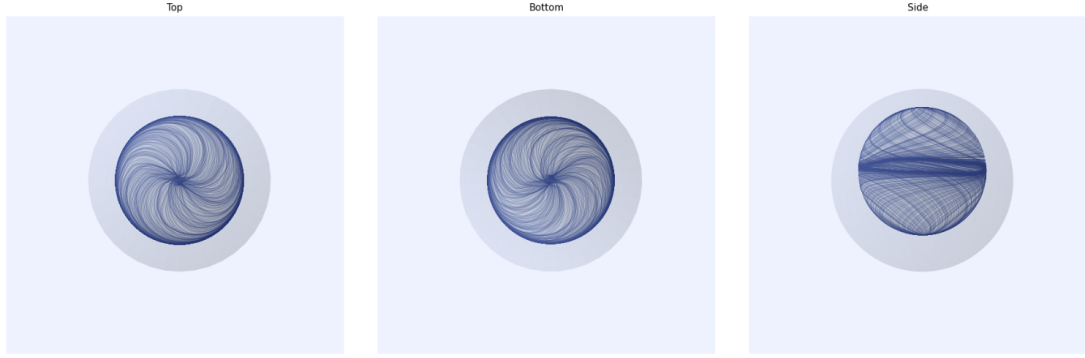


Figure 13: Results with the proposed two-scale feature method and ε -continuation. The sharp localization of defect cores and crisp director field streamlines demonstrate the effectiveness of the combined approach.

The computed director field exhibits the two $+1$ topological defects predicted by the Poincaré–Hopf theorem, positioned at locations that minimize the total free energy, with smooth and physically consistent streamline organization around the defect cores.

Beyond the visual structure of the director field, the EVNN scheme also preserves the energy-dissipation property of the underlying gradient flow on this problem. Figure 14 shows the discrete Landau–de Gennes free energy $\mathcal{F}[\tilde{\mathbf{Q}}]$ as a function of time for the proposed two-scale + ε -continuation method. The energy decreases monotonically from $\mathcal{F} \approx -8.85$ at the start of the simulation to $\mathcal{F} \approx -9.23$ at $t = 3$, with a smooth convex profile and no spurious oscillations. This confirms that the variational time-stepping scheme (5) enforces $\mathcal{F}[\tilde{\mathbf{Q}}^{n+1}] \leq \mathcal{F}[\tilde{\mathbf{Q}}^n]$ in practice on the full 3D LC shell problem, and not only on the simpler Allen–Cahn benchmarks.

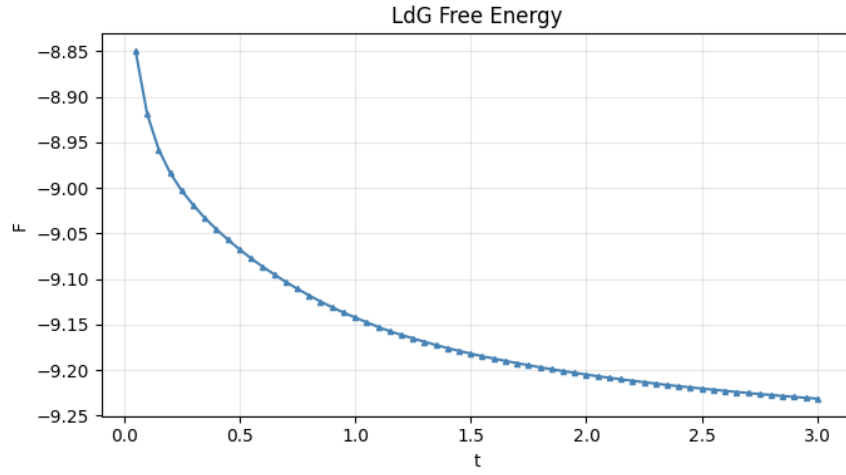


Figure 14: Discrete Landau–de Gennes free energy $\mathcal{F}[\tilde{\mathbf{Q}}]$ versus time for the proposed EVNN method with two-scale features and ε -continuation on the 3D liquid crystal shell. The energy decreases monotonically, confirming the energy-dissipation property of the EVNN scheme in this regime.

7 Conclusions and Future Work

7.1 Conclusions

This work develops and validates a comprehensive neural network approach to multiscale gradient flow problems with sharp interfaces, with application to liquid crystal shells. The key contributions are:

- **Two-scale feature embedding**, which augments network inputs with explicitly rescaled coordinates $x/\sqrt{\varepsilon}$ to resolve sharp $O(\varepsilon)$ features and reduce spectral bias;
- **ε -continuation training**, which progressively decreases the small parameter from large to target values, avoiding ill-conditioning and enabling convergence;
- **Energetic Variational Neural Networks (EVNN)**, which discretizes the gradient-flow structure variationally to guarantee energy dissipation at every time step.

Validation on one- and two-dimensional Allen–Cahn equations confirms that each method contributes meaningfully to improving solution quality. In the 1D setting, both Fourier feature and two-scale embeddings successfully resolve sharp interfaces, with EVNN guaranteeing energy dissipation throughout. In the 2D setting, the Fourier-feature EVNN achieves lower pointwise errors against the FEM reference, while the two-scale method with continuation trades pointwise accuracy for explicit ε -adaptivity. On the three-dimensional liquid crystal shell problem, where the Fourier-feature variant did not converge to a sharply resolved director field, the proposed combination of EVNN, two-scale features, and ε -continuation qualitatively reproduces topological defect structures with sharp localization and physically consistent director-field organization, while preserving the expected monotonic energy decay (Figure 14). This is a clear visual improvement over the baseline EVNN without two-scale features or continuation, though we note that quantitative benchmarking on the LC problem is left to future work.

7.2 Future Work

The developed techniques are general-purpose and directly applicable to any multiscale gradient-flow problem with sharp features and small parameters. Future di-

rections include:

- Extending to **higher-resolution benchmarks** with more sophisticated FEM reference solutions to enable quantitative error analysis for the LC problem;
- Applying these methods to other complex fluid systems, including those with **hydrodynamic coupling**;
- Exploring performance on **different geometries and parameter regimes**, including varying shell thickness and elastic anisotropy;
- Investigating **adaptive continuation schedules** that automatically tune the ε sequence based on training loss behavior.

Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this work, the author used AI tools to assist with language polishing, proofreading, and coding. The author takes full responsibility for the report.

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